

Exam Date & Time: 08-May-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VI Semester - End Semester Examination - April / May -2024

Bio-Informatics [CSE-4012]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Discuss in detail at least five characteristics of a Good Biological Database. List and interpret at least three features of the GenBank Nucleotide Database. (4)
 - A)
 - B) Enumerate and elaborate on the six steps that are adapted by the UniProtKB Database for automatic annotation. Bring out the significance and need of Biological databases which are highly specific to a specialized micro-domain. (4)
 - C) Determine the highest matched score for aligning GGTAAG and ATGTC using a simple matrix-driven alignment strategy. (2)
- 2) Distinguish between Global Alignment and Local Alignment. Illustrate the steps in the Smith-Waterman algorithm to justify Local Alignment between a pair of nucleotide bases. (4)
 - A)
 - B) Illustrate the working of DbClustal for progressive alignment and iterative alignment which involves two sets of iterations in the context of gene sequence alignment. (4)
 - C) In the context of bioinformatics analyses, interpret the need for Format Conversion by citing a tool as an example. (2)
- 3) State and interpret at least one exhaustive and one heuristic strategy for multiple sequence alignment of biological sequences. (4)
 - A)
 - B) Discuss the concepts of Maximum Parsimony and Bootstrapping along with the schematic representation of a bootstrap analysis for the construction and evaluation of phylogenetic trees. (4)
 - C) State and interpret the Shimodaira-Hasegawa test along with an equation in the context of guide trees. (2)

- 4) With an illustration using a flow diagram, discuss the overview of the MUSTANG algorithm. Also, give the configuration setup for the implementation of the MUSTANG algorithm by specifying the dependent factors. (5)
- A)
- B) With the help of necessary diagrammatic depictions, discuss the role of Bioinformatics in drug discovery focusing on its applications in drug metabolism, pharmacology, and toxicology. (5)
- 5) With an abstract architecture of Bio2RDF knowledge system framework, illustrate and elaborate on the integration of bioinformatics tools with associated knowledge via Ontological Design, Structured NameSpaces, and Triple Stores. Also, explain the need of Rdfizer programs for Semantic Integration of Bio information and Derived Knowledge. (5)
- A)
- B) With an illustration of the steps involved in RaptorX, discuss the protein structure prediction using the Threading Method. Also, elaborate on the results and interpretation for prediction of secondary and tertiary protein structures. List the steps adapted by Open PSIPRED for secondary protein structure prediction. (5)

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